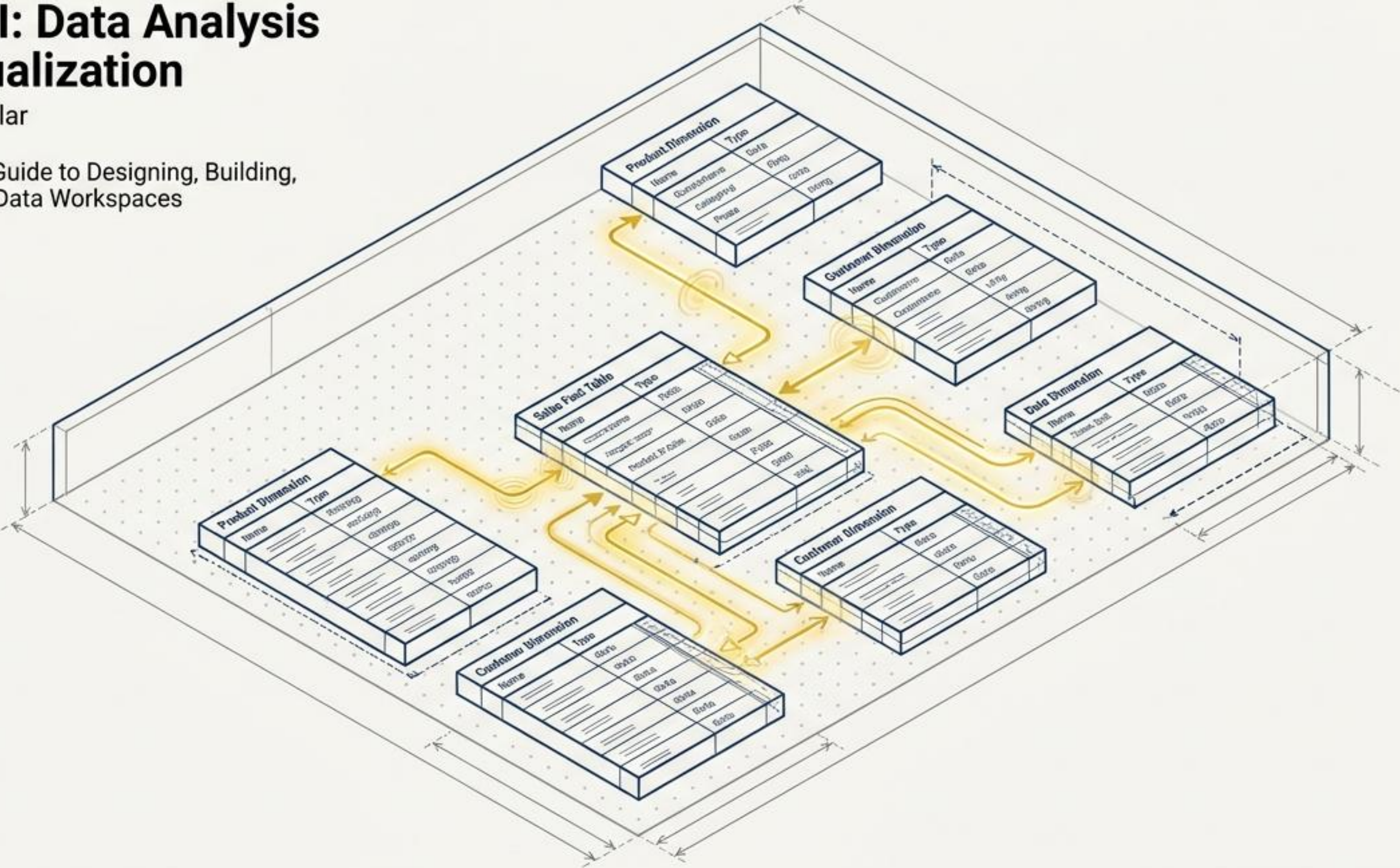


Power BI: Data Analysis and Visualization

dr Zbigniew Galar

The Architect's Guide to Designing, Building, and Optimizing Data Workspaces



Defining the Deliverable: Reports vs. Dashboards

A dashboard provides a graphical summary for business overviews. While users often use the terms interchangeably, the Power BI ecosystem structurally separates the two.

Reports



- **Created in:** Power BI Desktop or Service
- **Structure:** Multiple pages
- **Foundation:** Must be underpinned by a data model
- **Content limit:** Cannot contain streaming content

Dashboards

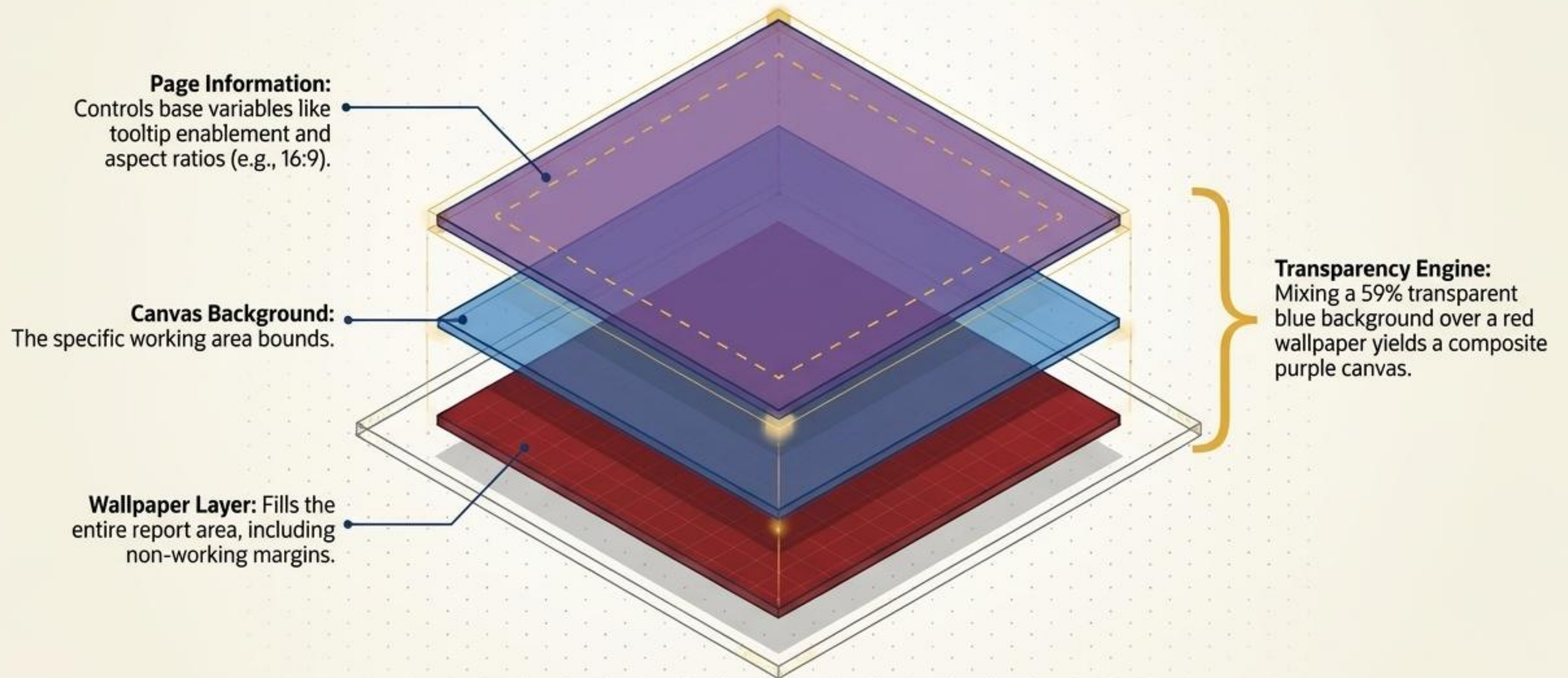


- **Created in:** Power BI Service only
- **Structure:** One virtual page
- **Foundation:** Uses visuals pinned from published reports
- **Content limit:** Can contain streaming content

Both formats can be shared via organizational apps.

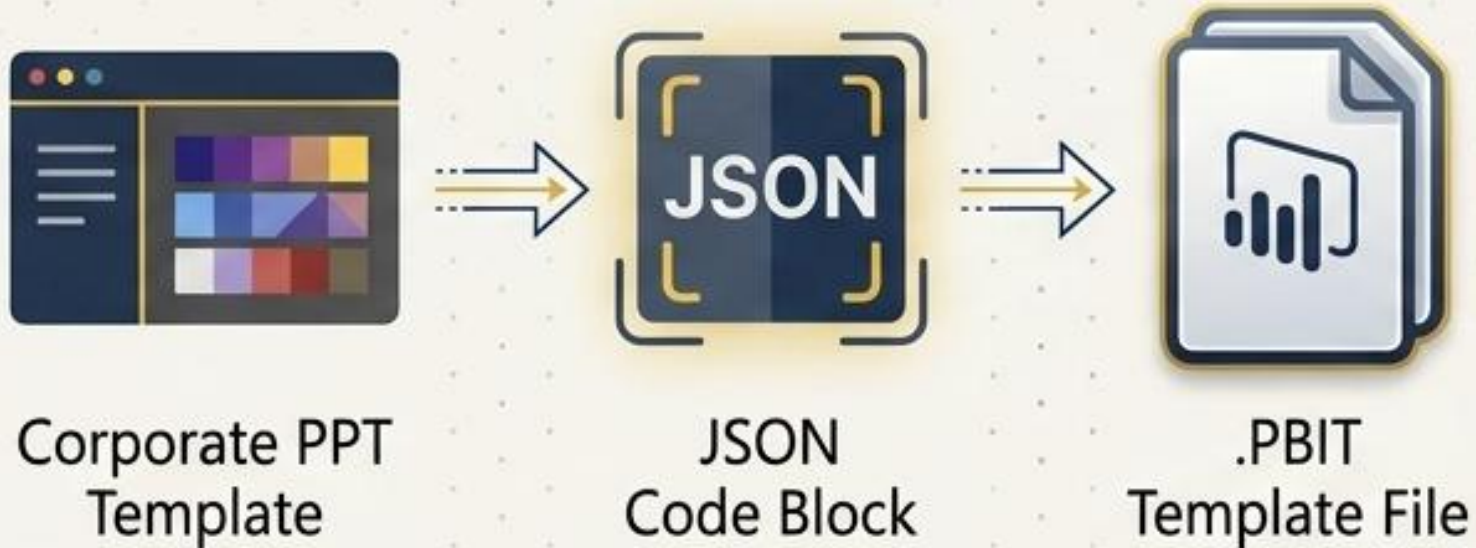
The Anatomy of the Canvas

Before dragging a single field, define the spatial parameters of the report. The canvas utilizes overlapping layers that react dynamically through transparency settings.



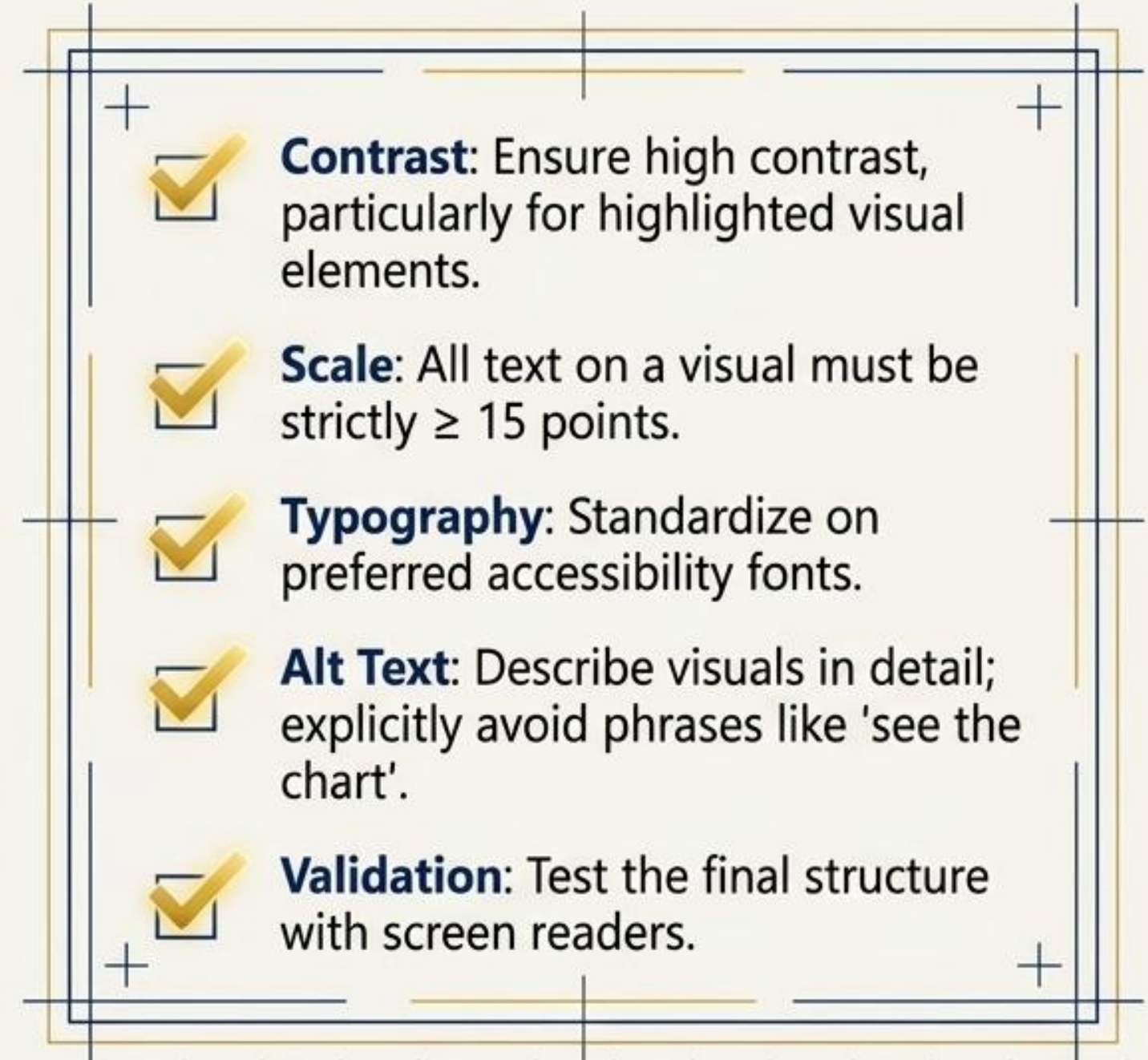
Standardization and Accessibility

The Blueprint - PBIT



- ✓ Avoid the design rabbit hole. Extract colors and images from corporate PowerPoint templates.
- ✓ Encode brand standards using JSON.
- ✓ Distribute as a .PBIT (Power BI Template) file to ensure all new .PBIX files start perfectly branded.

Universal Access



The Visualization Taxonomy

Power BI offers 37 standard visual types. The geometry of the visual must be strictly dictated by the shape and intent of the data.

Comparison

Clustered column, bar, line



Change over time

Line, area



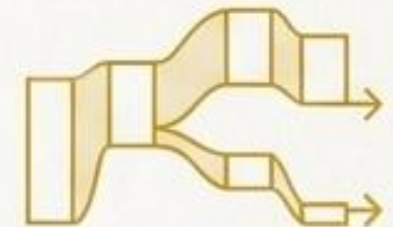
Parts of a whole

Stacked column/bar, pie, donut, treemap



Flow

Area, specialty visuals



Spatial

Globe, filled maps, ArcGIS



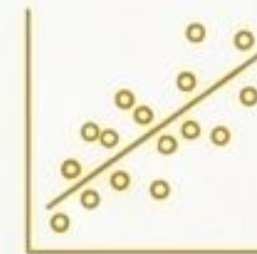
Distribution

Line, scatter, box & whiskers



Correlation

Scatter, line



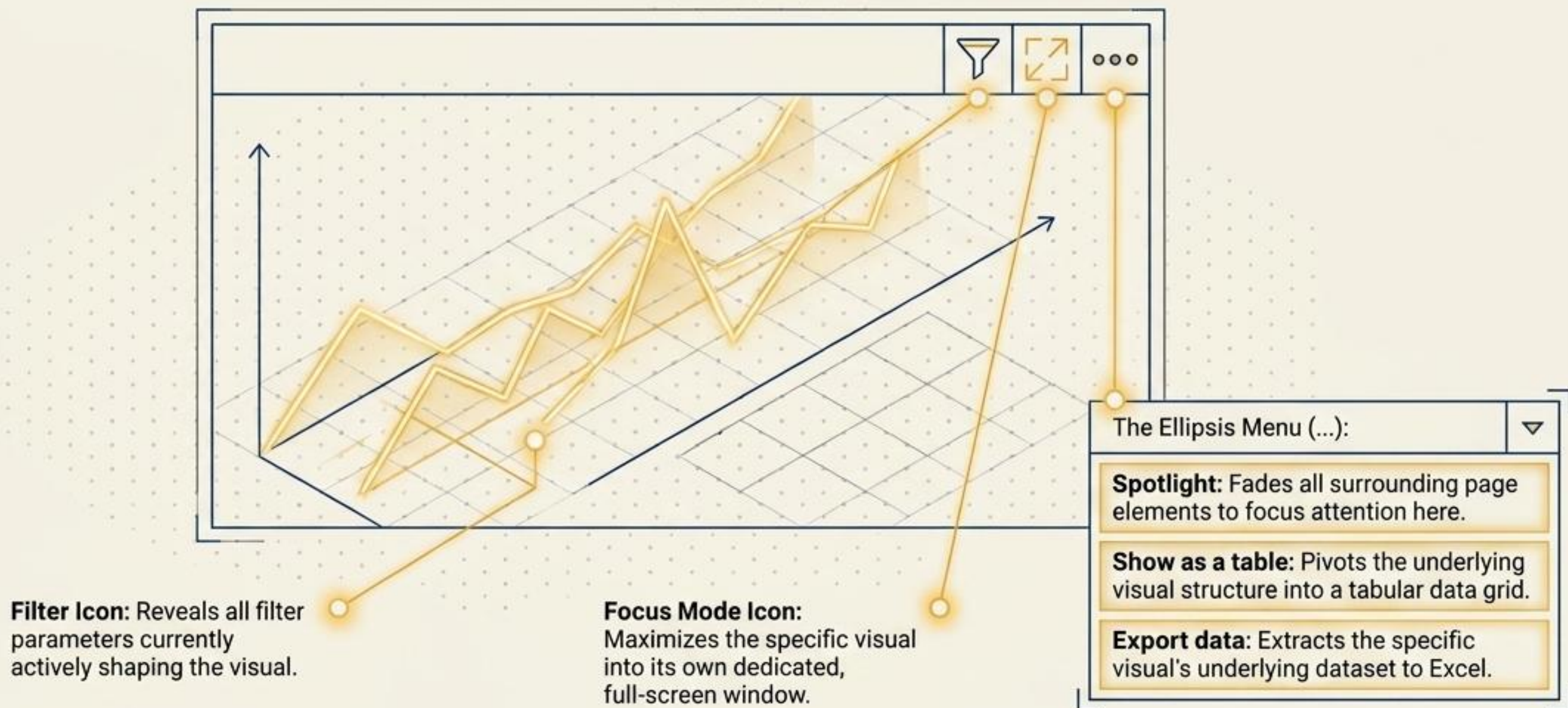
Single

Cards, KPI, gauge



The Anatomy of a Visual

Regardless of the chosen geometry, every visualization shares a standard set of header controls designed for user interaction and data extraction.



Formatting Paradigms: Visual vs. General

The Format pane is divided into two strict hierarchies. Always explore standard formatting thoroughly before resorting to custom visual solutions.

Visual Tab (Geometry-Specific)



The Visual Tab UI mockup shows a header with a title bar, a search bar, and a dropdown menu. Below the header, the word "Visual" is centered and underlined. The main content area contains the text "Controls features unique to the specific chart type." followed by a bulleted list of features.

Visual

Controls features unique to the specific chart type.

- X/Y Axis scales
- Legend positioning
- Column subtotals
- Cell elements

General Tab (Universal Properties)



The General Tab UI mockup shows a header with a title bar, a search bar, and a dropdown menu. Below the header, the word "General" is centered and underlined. The main content area contains the text "Controls features common to every object on the canvas." followed by a bulleted list of features.

General

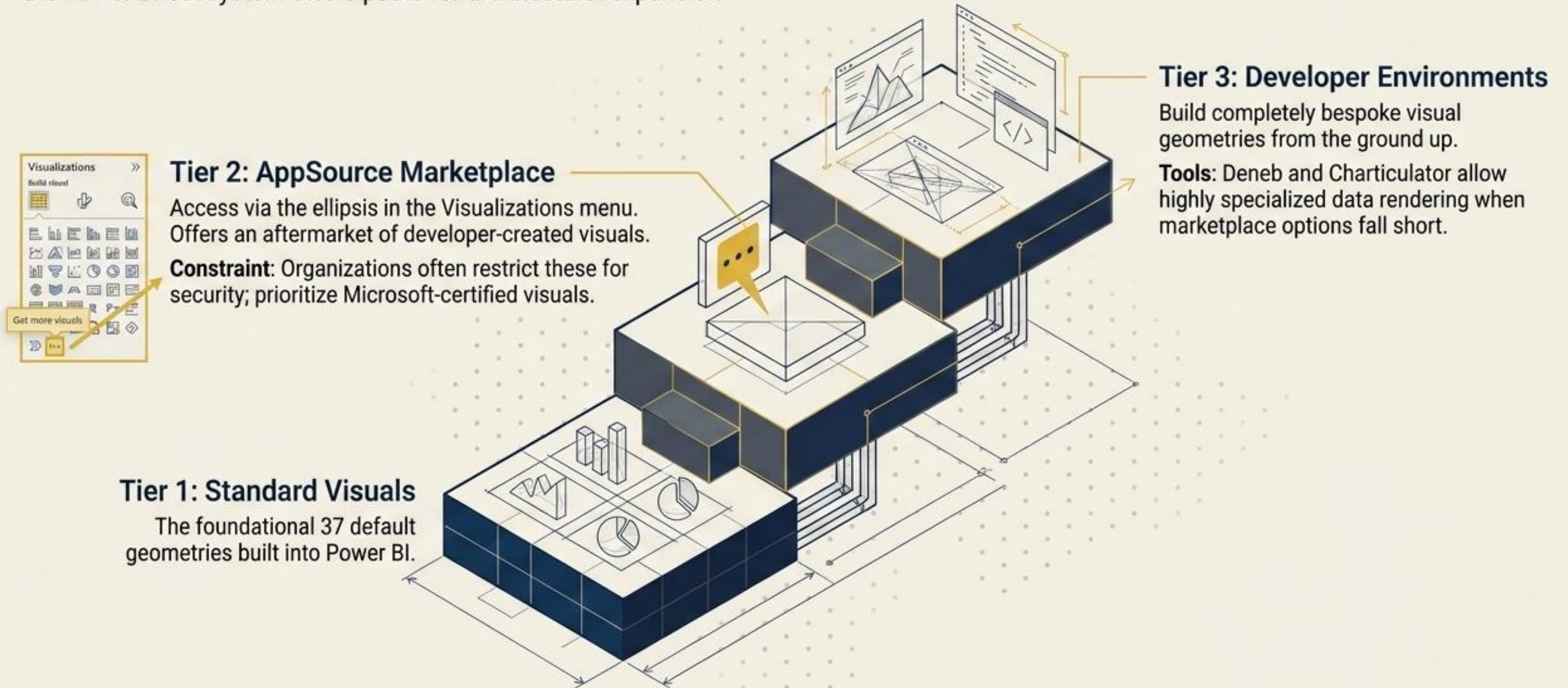
Controls features common to every object on the canvas.

- Title text
- Background opacity
- Borders
- Drop shadows
- Tooltip enablement



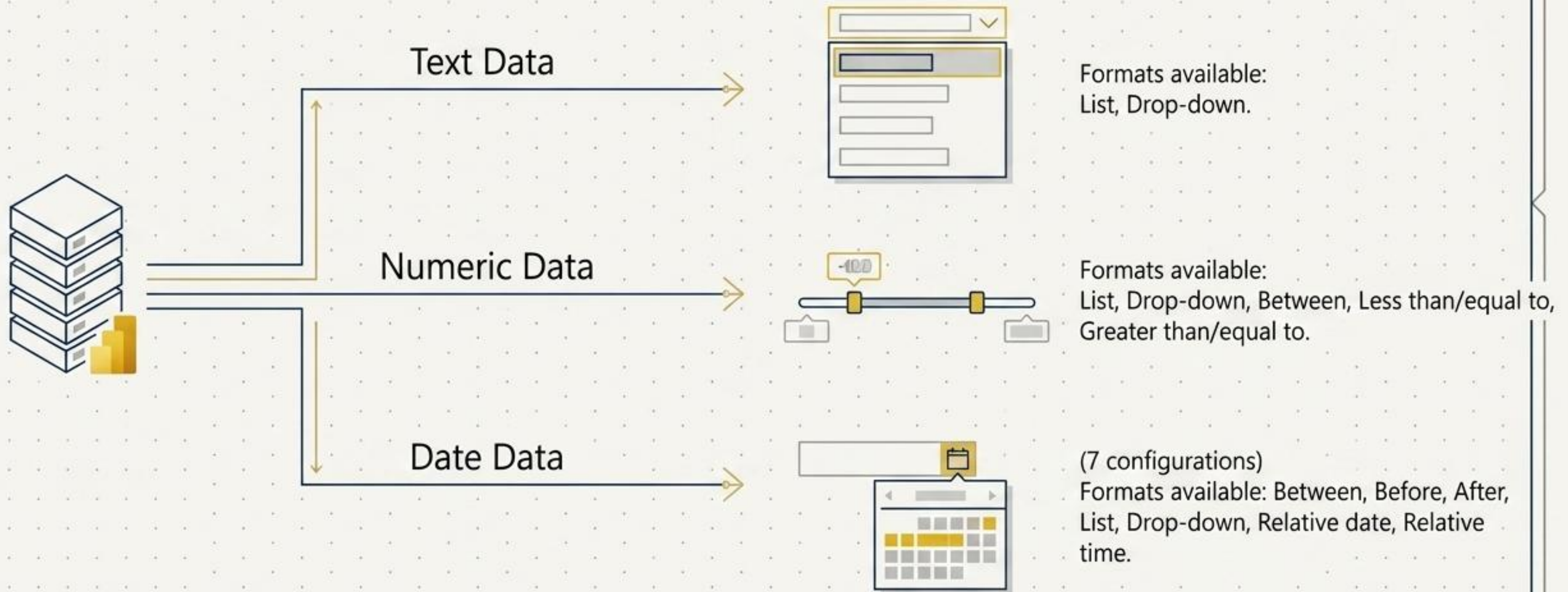
Pushing the Boundaries: Custom Visuals

When standard geometries cannot fulfill specialized analytical requirements, the Power BI ecosystem offers paths for architectural expansion.



Architecting Interactivity: The Slicer Toolkit

Slicers are a specialized visual category used to hand filtering control directly to the user. The format options adapt intelligently to the underlying data type.

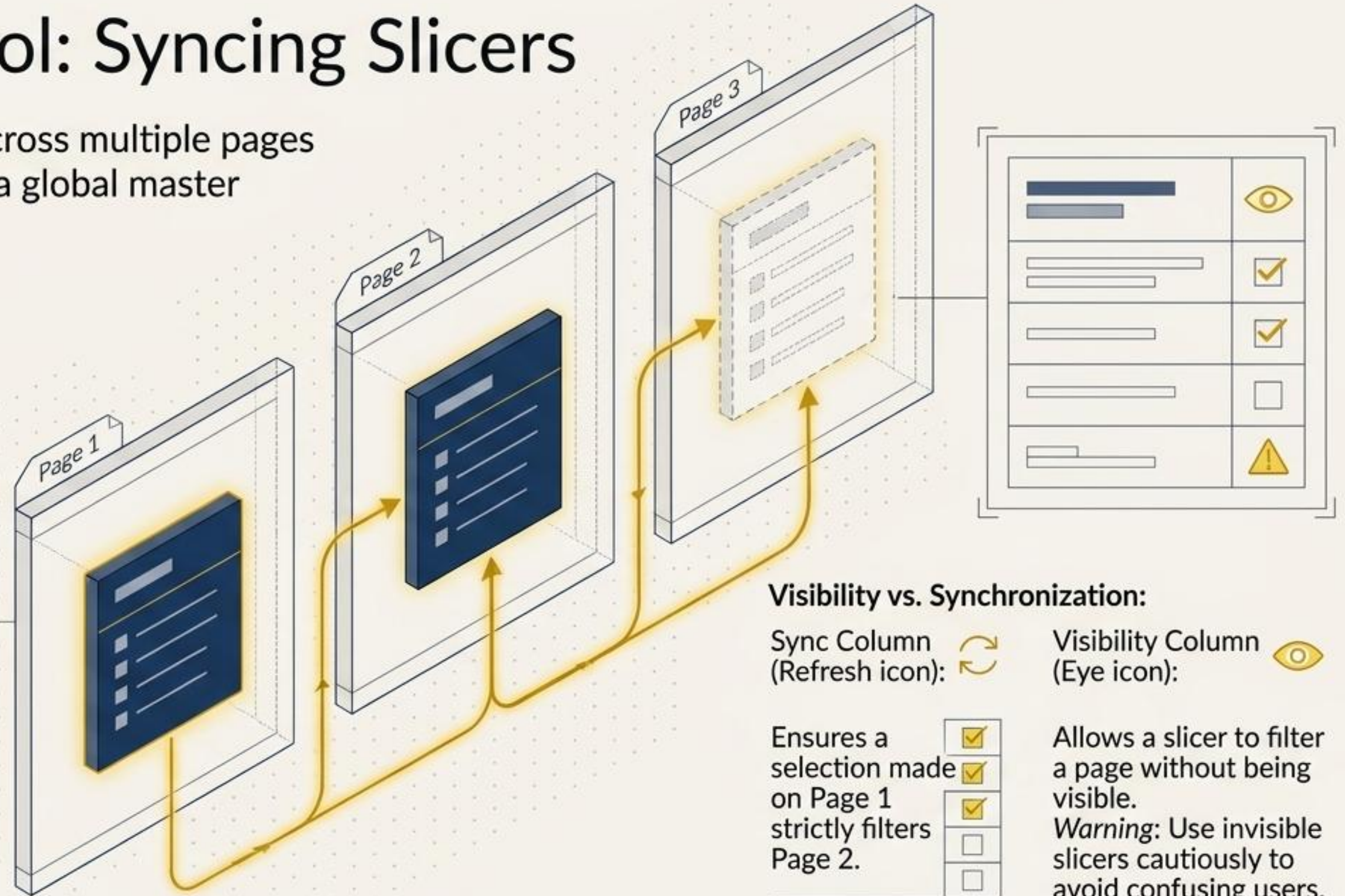


Global Control: Syncing Slicers

Slicers can dictate logic across multiple pages simultaneously, acting as a global master switch for the report.

The Sync Mechanism:

Copying a slicer (Ctrl+C) triggers an automatic sync prompt. The Sync Slicers Panel provides fine-tuned control over the global ecosystem.



Visibility vs. Synchronization:

Sync Column (Refresh icon): 

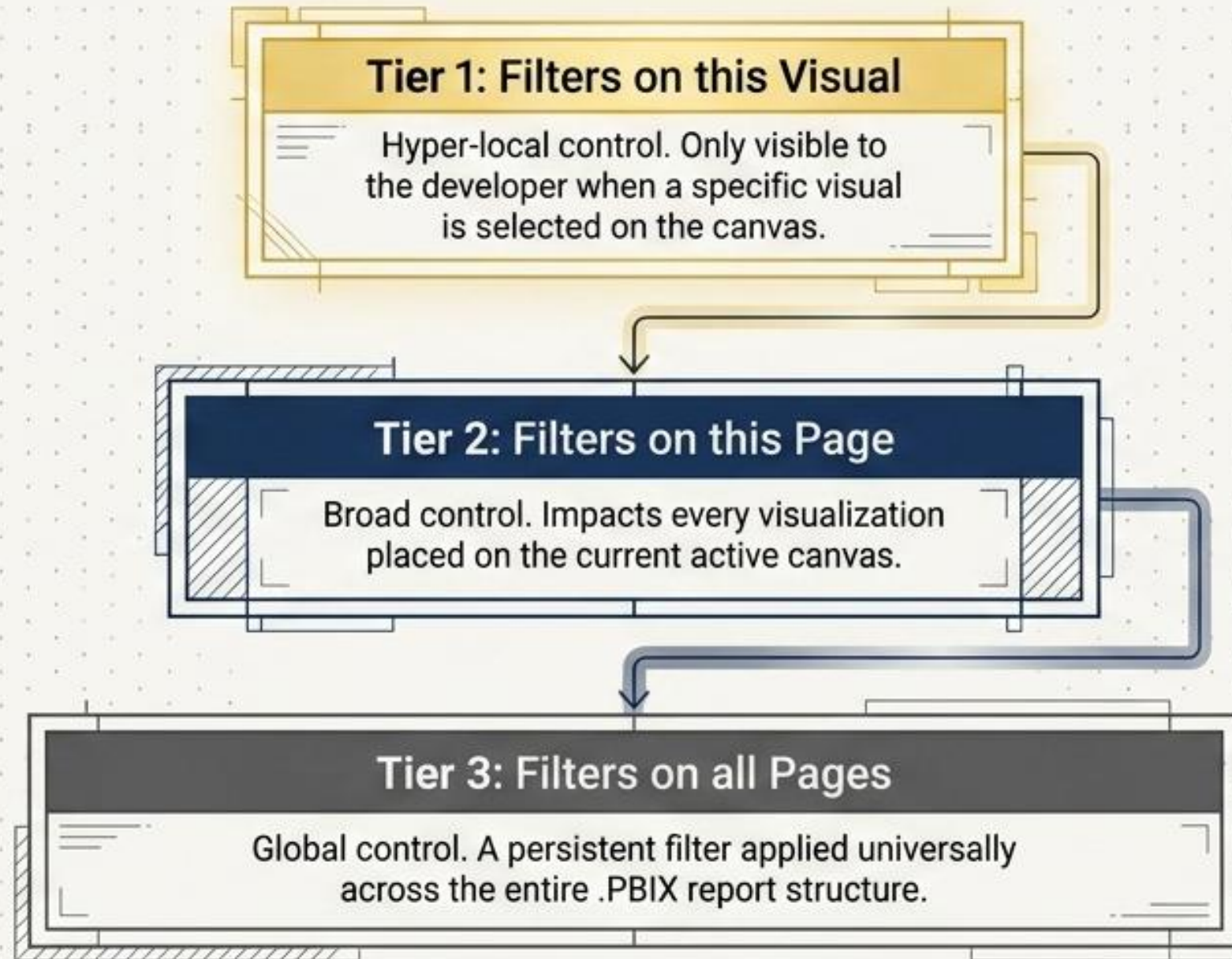
Ensures a selection made on Page 1 strictly filters Page 2.

Visibility Column (Eye icon): 

Allows a slicer to filter a page without being visible.
Warning: Use invisible slicers cautiously to avoid confusing users.

Spatial Hierarchy: The Filters Pane

While slicers consume valuable canvas real estate and have functional limits, the Filters Pane offers a dedicated, hierarchical engine for report filtration.



Advanced Filtering Logic and UX Protection

Dragging a field into the Filters Pane unlocks complex logical conditions and allows developers to protect the user experience from accidental modification.

DIAGNOSTIC

Three Modes of Execution:

- **Basic Filtering:** Operates identically to standard slicer checkbox behavior.
- **Advanced Filtering:** Enables combined conditional logic using strict "And/Or" parameters (e.g., contains "X" AND does not contain "Y").
- **Top N Filtering:** Exclusive to visual-level filters. Mathematically restricts the visual to the highest or lowest "N" data points based on a specified value metric.

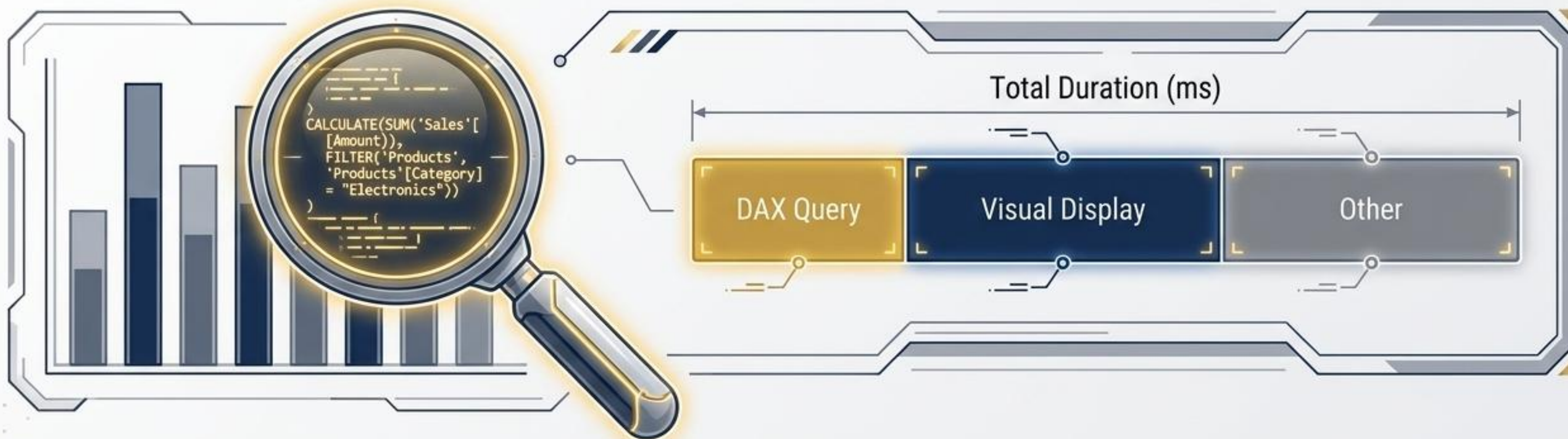
UX Governance:



Developers can **Lock** a filter to prevent user modification, or **Hide** the filter card entirely to streamline the user interface.

The Hidden Engine: Performance Analyzer

The first rule of performance: Every visual creates a DAX expression. Even if you never write a line of code, your report is executing it.



Measuring Impact:

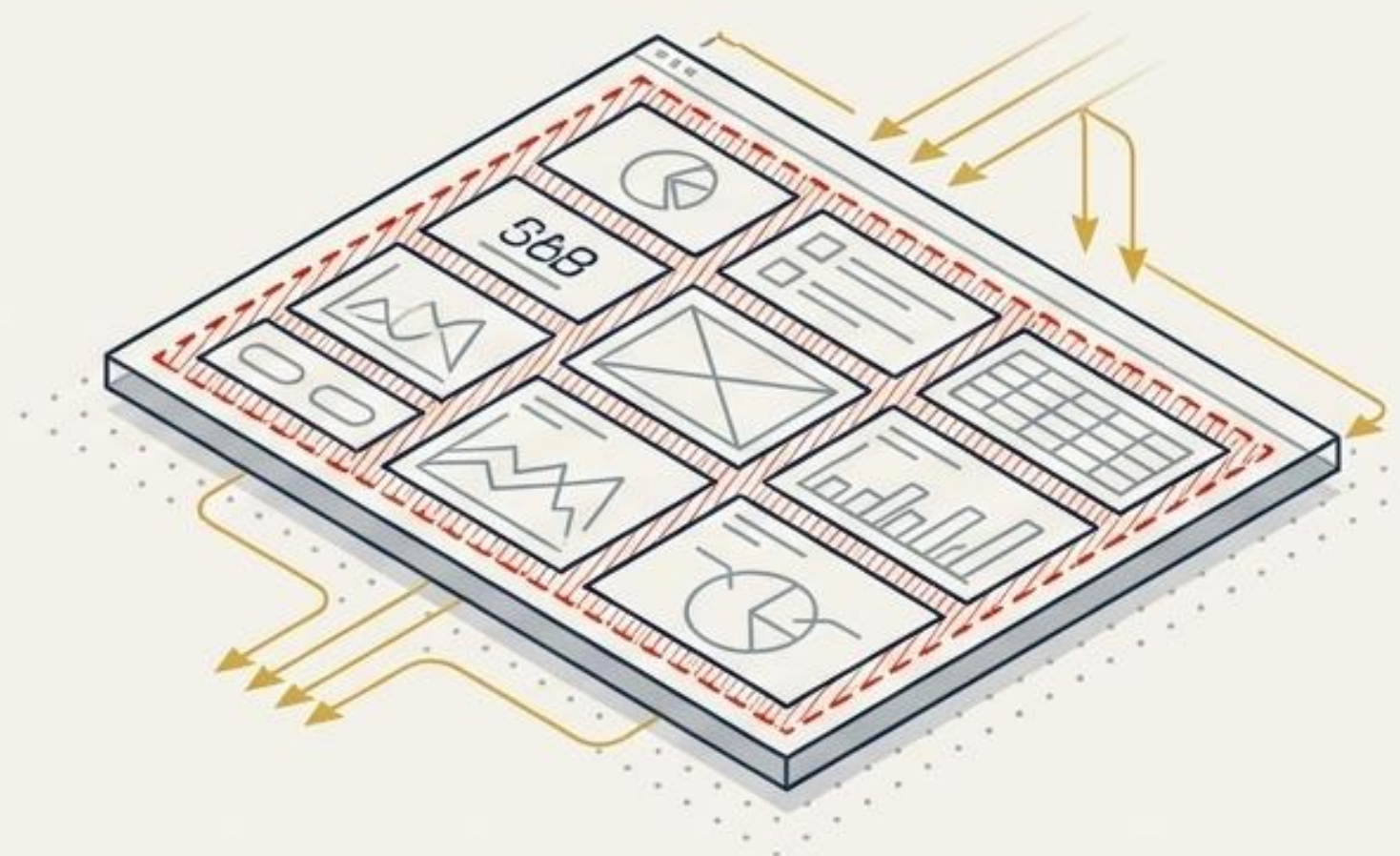
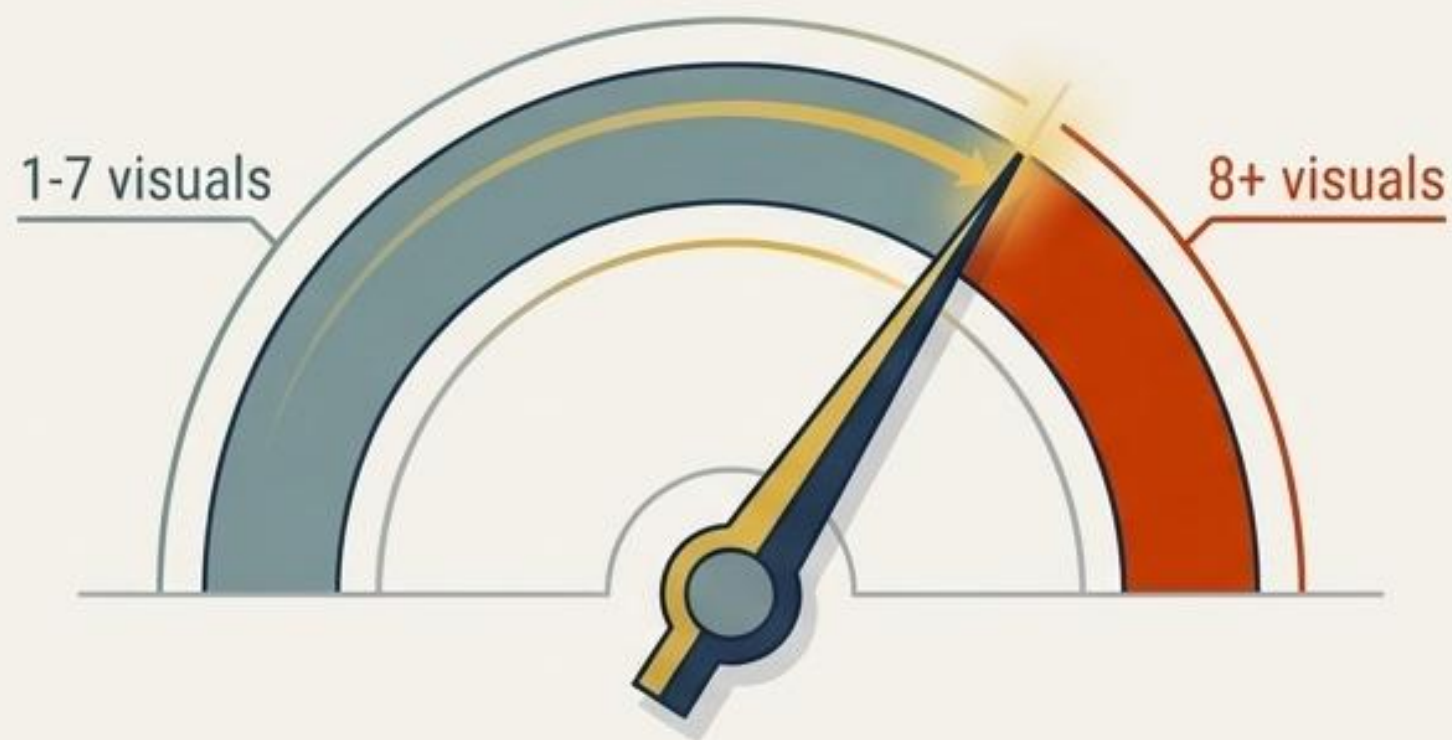
Found on the View tab, the Performance Analyzer records user interactions and measures render duration in milliseconds. It isolates bottlenecks by splitting rendering time into DAX query, Visual display, and Other.

The Developer's Tool:

Copy the raw DAX queries generated by visuals to study syntax logic. If a visual utilizing custom measures is slow, you must optimize by rewriting the underlying DAX formulation.

Cumulative Impact: The 5-to-7 Rule

Perfect is the enemy of done when writing DAX, but structural bloat will inevitably kill report responsiveness.



● **The Golden Ratio:** There is no strict physical limit, but the definitive rule of thumb is to aim for a maximum of 5 to 7 visuals per page.

● Every card, button, and slicer requires processing time. Performance degrades due to the **cumulative impact** of concurrent DAX queries firing simultaneously upon page load.

● **Validation:** Use the Performance Analyzer's 'Clear' then 'Refresh Visuals' function to measure the total, cumulative responsiveness of an entire page layout.

The Optimization Matrix

A beautifully designed canvas cannot mask a poorly engineered data model.
Resolve these fundamental structural flaws before publishing.

	High Cardinality		Separate date and time into two distinct columns in Power Query.
	Unnecessary Columns		Ruthlessly remove unused data fields at the source.
	Many-to-Many Relationships		Engineer bridge tables to resolve ambiguity.
	Bidirectional Relationships		Reverse relationships dynamically using DAX (e.g., CROSSFILTER).
	Calculated Columns		Replace with measures, or push the calculation upstream to Power Query.
	Excessive Interactions/Tooltips		Disable cross-filtering interactions if unneeded; disable tooltips to prevent silent, secondary DAX queries.

*"Perfect is the enemy of done—don't let the quest for perfect DAX get in the way of writing DAX.
Good DAX takes practice." — dr Zbigniew Galar*